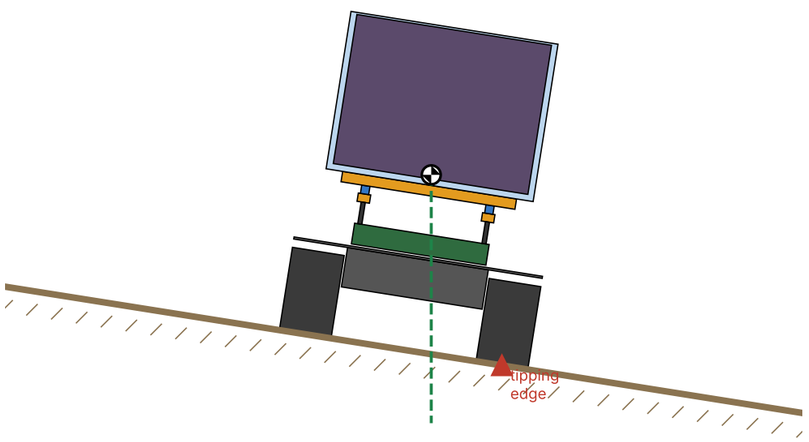
BINBOT — STABILITY LIMITS WITH A FULL BIN (500 kg olives, deck raised)

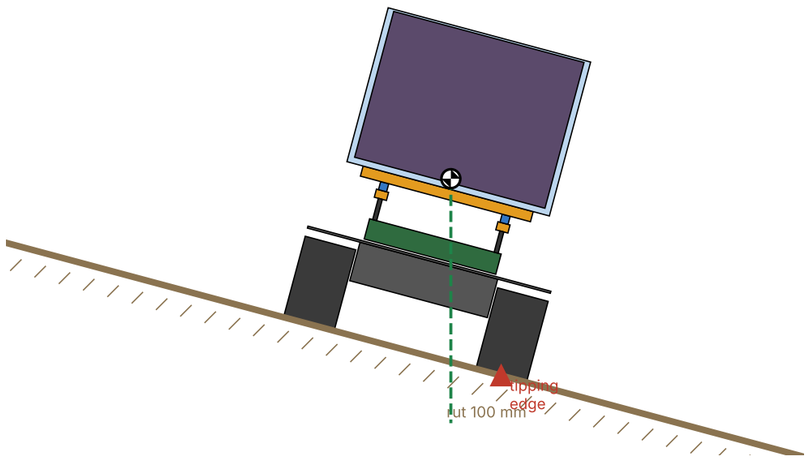
A CROSS-SLOPE LIMIT 9° (driving)

OK: CoG well inside.
Alarm 7°, auto-stop 12°.



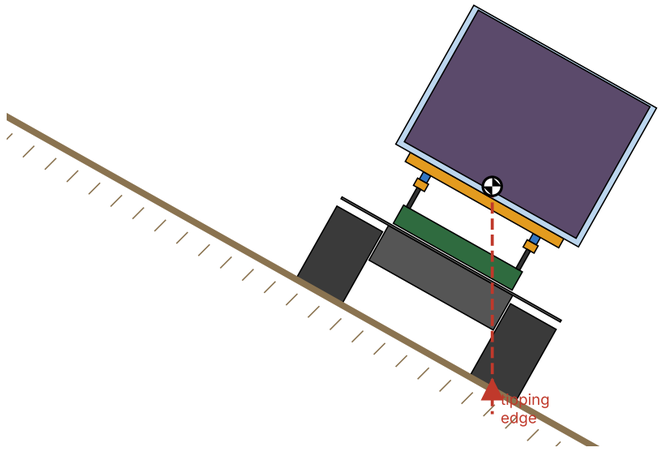
B ROAD: 9° CAMBER + 100 mm RUT = 15°

OK: still 14° from tipping.
Ruts are why the limit is far below the tip angle.



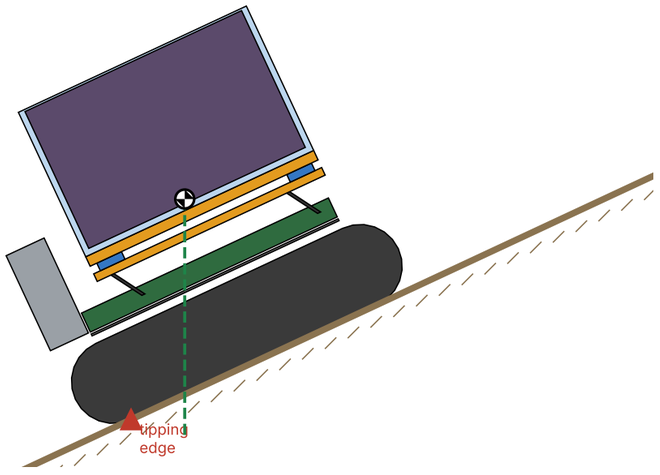
C SIDEWAYS TIP POINT 29° — NEVER

CoG exactly over the edge:
the robot rolls over.



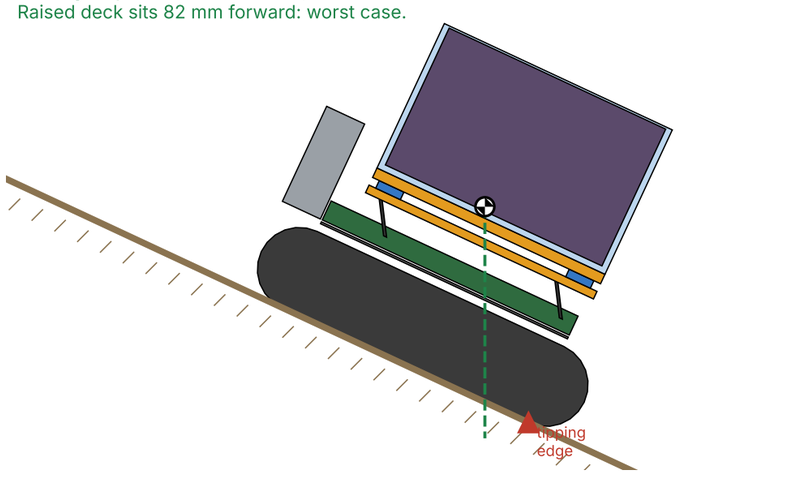
D CLIMBING 25° (front up)

OK: tips backwards only at 39°.



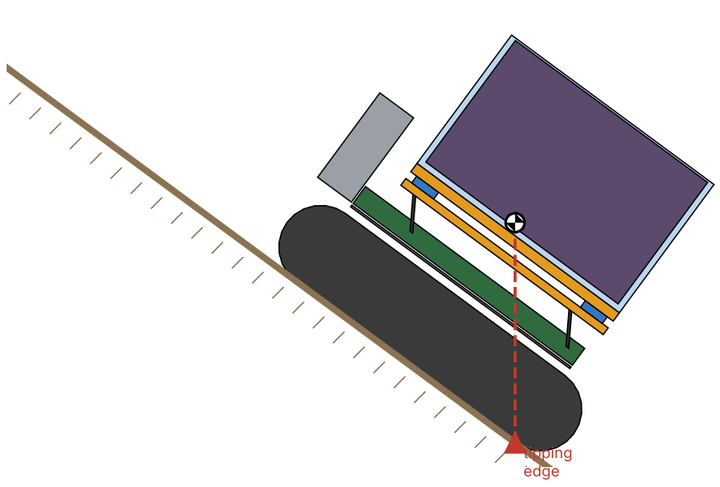
E GOING DOWN 25° (front down)

OK: tips forwards at 36°.
Brake gently (0.5 m/s² ≈ 3° extra).
Raised deck sits 82 mm forward: worst case.



F FORWARD TIP POINT 36° — NEVER

CoG over the front edge
of the track contact.



Load state	CoG height	Tip sideways	Tip fwd / back	Cross-slope limit	Up/down limit
Full bin 500 kg, deck RAISED (normal)	845 mm	29°	36° / 39°	9°	25°
Full bin 500 kg, deck LOWERED (road mode)	790 mm	31°	40° / 39°	9°	25°
Empty bin	457 mm	46°	56° / 54°	14°	25°
No bin	393 mm	50°	60° / 57°	15°	25°

ROAD MODE (recommended)
On roads and long transfers, carry the full bin with the deck LOWERED:
CoG drops 55 mm, sideways tip 29° → 31°.
Needs 8 kN actuators (instead of 6 kN) to lift a full bin
again before entering a FULL bay. +€60–100.

Masses are estimates: weigh the robot once built and
re-run cad/stability.py; the software limits use these values.

Operating limits: sideways ≈ 30 % of the tip angle (ruts, bumps, turning, soft ground); up/down = tip angle − 11°
(3° braking + 8° bumps), capped at 20° / 25°. Sideways tipping edge taken at the track centre line (soft ground).

DWG 004 REV C stability 2026-10-28
PRELIMINARY — masses estimated, base data assumed